



Figure 2: Packet-spoofing attack

look at internal traffic.

ARP spoofing: Since it is all about modifying or forging packets, modern attackers don't restrict themselves to IP fields. In this type of attack, which is also known as ARP poisoning, the attackers send spoofed ARP packets on the local area network, to associate the attackers' own machine's MAC address with the IP address of another host, which is the target. Due to this, traffic for the target now reaches the attackers' machine, due to the binding between IP and MAC addresses. Thus this becomes a local man-in-the-middle attack. The attackers can further hide by forwarding received data to the actual destination, and since there is no data lost in this process, it is almost impossible to find the man-in-the-middle stealing the data.

DNS spoofing: Also known as a DNS poisoning attack, this is more serious. In this case, the domain name system server is spoofed to alter entries of domain names to reflect the attackers' IP address. This results in sending Web or email traffic to the attackers' machine. This attack is achieved by creating multiple forged packets wherein the IP, port and service type entries are modified to serve the purpose. The repercussions of such an attack are very dangerous, whereby a website can be defaced, or email data can be stolen.

Email and Web spoofing: Speaking about Layer 7, the same technique of impersonation can be stretched to crafting fake email addresses, Web requests and hyperlinks. This is usually done to plant a Trojan or a virus and spread it across. Please remember that spoofing is a basic component in planting a denial-of-service attack, mainly because the attacker would always want to hide behind a curtain.

Protecting FOSS systems

While the spoofing attack is a difficult one to tackle, there are a few preventive mechanisms that all network administrators should adopt in their infrastructure. Since the spoofing starts at Layer 2, the real protection has to be implemented in the critical network components such as routers, firewalls, switches, etc. Deploying the latest stateful firewall, and enabling features in it that

fight against source packet spoofing, can be a first level of defence. As a daily chore, network logs from firewalls, routers and switches can be parsed using a script, to see if multiple duplicate ACK packets are being generated. Also, if an unusual number of SYN packets are being observed, which are not being responded to, that could be a clue to spoofing. The real spoofing detection is to check a bunch of packets for a given set of source and destination IP addresses, and see that there is no other IP address involved in that communication session. This is a difficult task for a script program to achieve, and for complex networks, deploying an intrusion detection device usually helps to a great extent. There is a thumb rule while inspecting packets, which says that in a network packet log, if an internal LAN IP address is being shown in the log file capturing data for an external interface, that indicates there is a problem.

Linux FOSS systems come with a built-in, but usually less-known feature called source address verification. It is a kernel feature which, when turned on, starts dropping packets that appear to be arriving from the internal network, but in reality are not. Most of the latest kernels, such as Ubuntu and CentOS, do support it, but if your Linux distro does not, it is time to upgrade. Modifying the `hosts.conf` file to add '`nospoof on`' is another level of defence to try.

In terms of detection, for smaller Linux networks, a nice utility called `arpwatch` is very useful. This keeps track of MAC and IP addresses, records all changes, and can be scripted to alert administrators about a possible attack. Scripting can also be done to go through network interface logs and look for anomalies in terms of source address forging. 

Summary

Packet spoofing is a difficult type of attack to tackle. It can result in serious data loss, and there are ways to detect it and stop it. Configuring firewalls, switches and routers is an important step to prevent networks from spoofing, and network administrators should know about it. Finance firms are found to be common victims to this kind of attack, and their IT management teams should take the necessary steps to prevent financial losses or damage to their reputation. Implementing IPS devices certainly helps in getting control over the IT network infrastructure security.

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